Gradients with a numerical check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Gradients with a numerical check before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Slopes and local change

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Parameters and architecture

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Activation functions

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Use one example with **input** x = 2 and target t = 3. The model predicts y = wx + b, and loss L(w, b) = (2w + b - 3) squared. At w = 1, b = 0, **prediction** is 2, residual is -1 and loss is 1.

There are two parameters in order [w, b].

To estimate the partial slope for w, change w only and keep b = 0. To estimate the partial slope for b, change b only and keep w = 1.

Moving both at once measures change along a combined direction; it does not isolate either **partial derivative**.

Perturbation from [1, 0]	Loss at minus step	Loss at plus step	Symmetric slope
w only with step 0.1	1.44	0.64	-4
b only with step 0.1	1.21	0.81	-2

The **gradient** is [-4, -2] in order [w, b]. Both entries refer to the same original parameter vector, not to a partially updated model.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Gradients with a numerical check trace and record the first step you can explain without help.

Use W2: Catch a denominator error for **Gradient**s with a numerical check. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Gradients with a numerical check explanation against the supplied facts before discussing it.

Use the worked case for **Gradient**s with a numerical check. Proposed explanation: “The **gradient** is one scalar”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Check a new example independently

Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Recover a gradient

> Recover a gradient. Apply the supplied Gradients with a numerical check facts and keep your reasoning.

For the lesson model at w = 1 and b = 1, calculate **prediction**, residual, loss and **gradient**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Catch a denominator error

> Catch a denominator error. Apply the supplied Gradients with a numerical check facts and keep your reasoning.

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Separate a direction from a coordinate

> Separate a direction from a coordinate. Apply the supplied Gradients with a numerical check facts and keep your reasoning.

Starting at [w, b] = [1, 0], both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either **partial derivative**.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Check a new example independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Check a new example independently. Apply the supplied Gradients with a numerical check facts and keep your reasoning.

Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Recognise a corner

> Recognise a corner. Apply the supplied Gradients with a numerical check facts and keep your reasoning.

Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Gradients with a numerical check to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled **gradient**s with a numerical check **evidence** sheet to Image-pattern Network Investigation. Show how you can estimate one **partial derivative** while other parameters stay fixed.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
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> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Gradients with a numerical check

> Check each coordinate of a gradient while holding the other parameter fixed. Explain the labelled step before checking the changed condition.

Check each coordinate of a **gradient** while holding the other parameter fixed.

Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/10

Gradients with a numerical check

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check w direction

Analytic derivative 2 x residual x 2=-4.



2. Check b direction

Analytic derivative 2 x residual=-2.



3. Numerical check

With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

CHANGE ONE THING

Swap the two gradient components in the program output.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check each coordinate of a **gradient** while holding the other parameter fixed.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

1. Numerical check

2. Check b direction

3. Check w direction

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Numerical check With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. A **gradient** lists the slope for each parameter, in a declared order.

Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Swap the two **gradient** components in the program **output**. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The coordinate-wise numerical check exposes the mismatch.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The coordinate-wise numerical check exposes the mismatch. The w perturbation changes y twice as strongly as the same b perturbation.

Repair a missing building block

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network’s final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

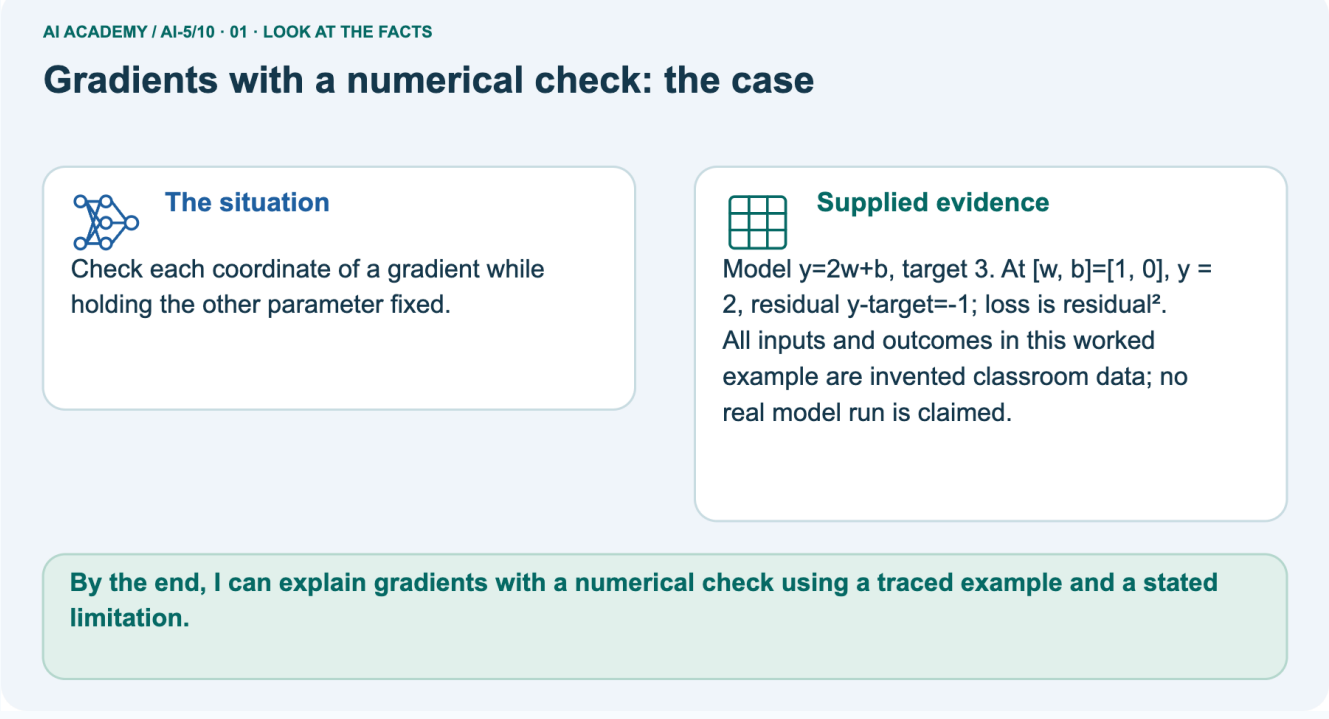
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Gradients with a numerical check picture from its labels, then record a relevant limit.

01. Gradients with a numerical check: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

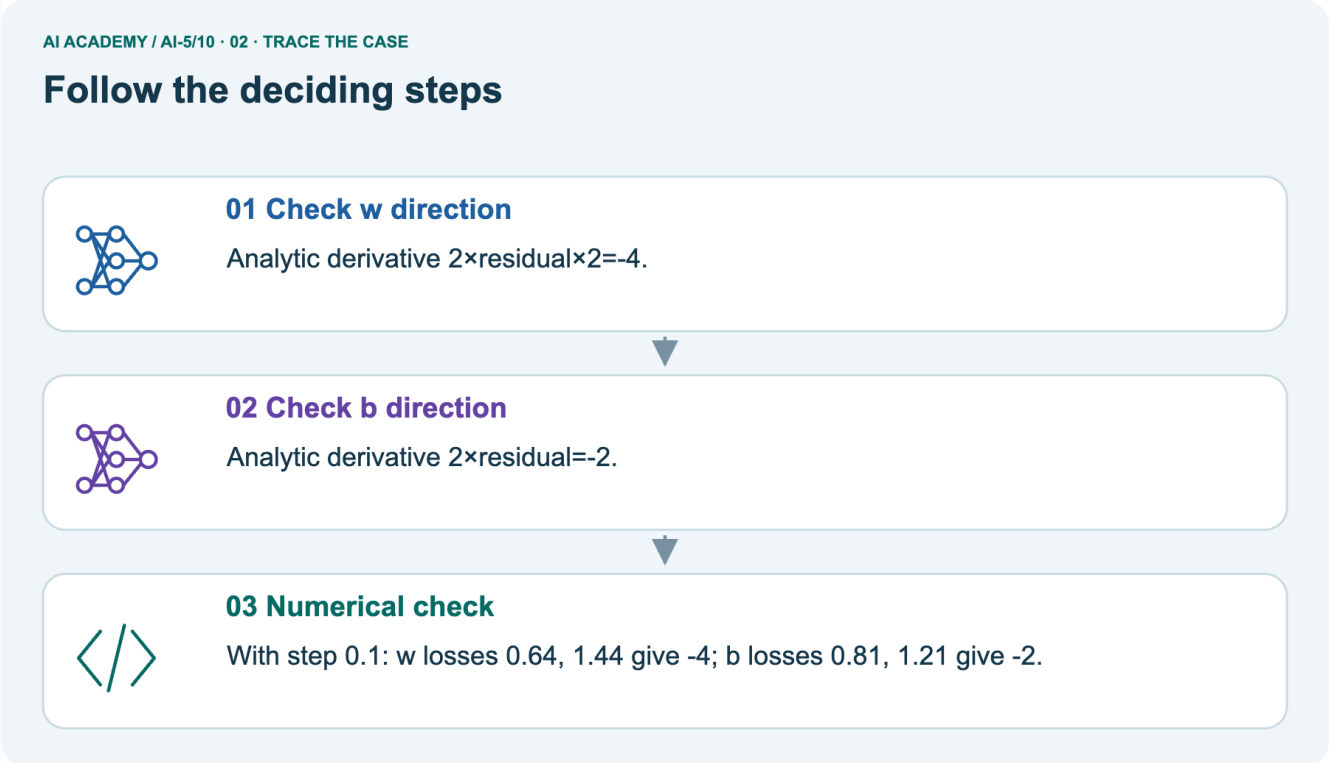
Check each coordinate of a **gradient** while holding the other parameter fixed. Supplied facts: Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



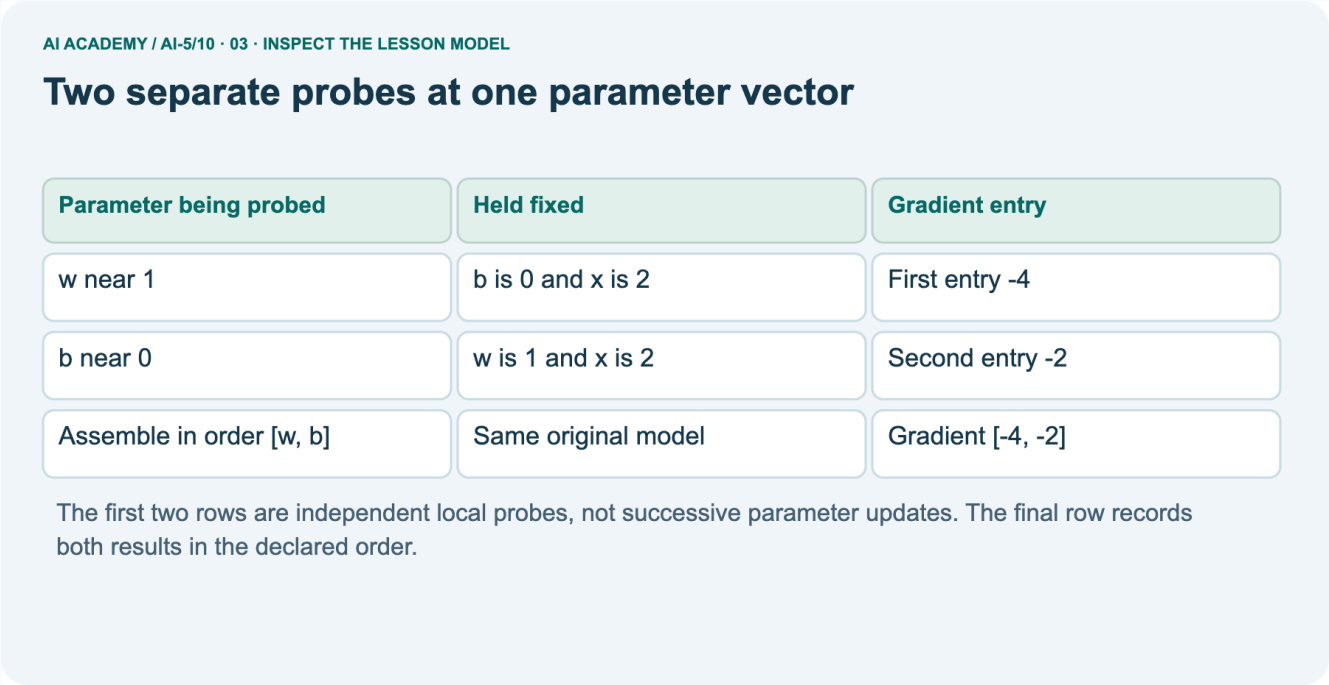
Follow the labelled picture with the matching explanation. Open the vector diagram

Check w direction: Analytic derivative 2×residual×2=-4. → Check b direction: Analytic derivative 2×residual=-2. → Numerical check: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two separate probes at one parameter vector



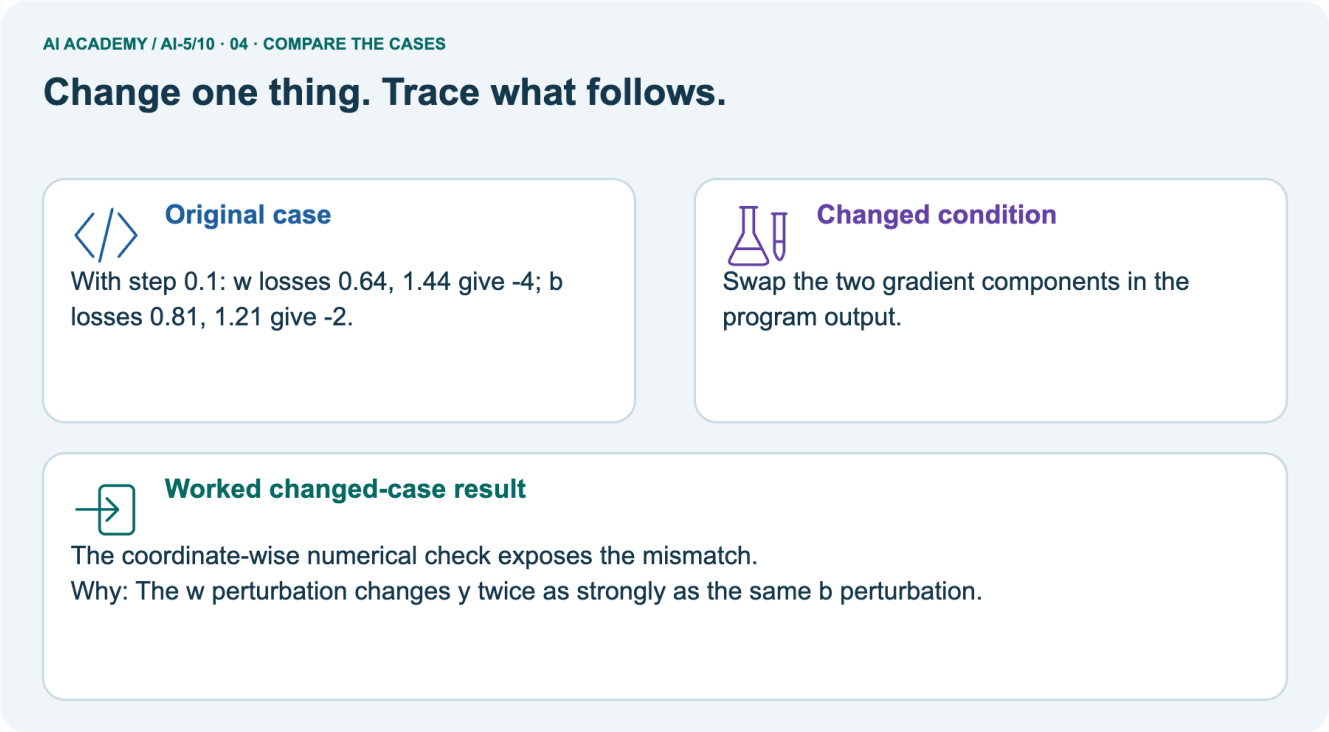
Follow the labelled picture with the matching explanation. Open the vector diagram

Two separate probes at one parameter vector. w near 1 | b is 0 and x is 2 | First entry -4. b near 0 | w is 1 and x is 2 | Second entry -2. Assemble in order [w, b] | Same original model | **Gradient** [-4, -2]

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

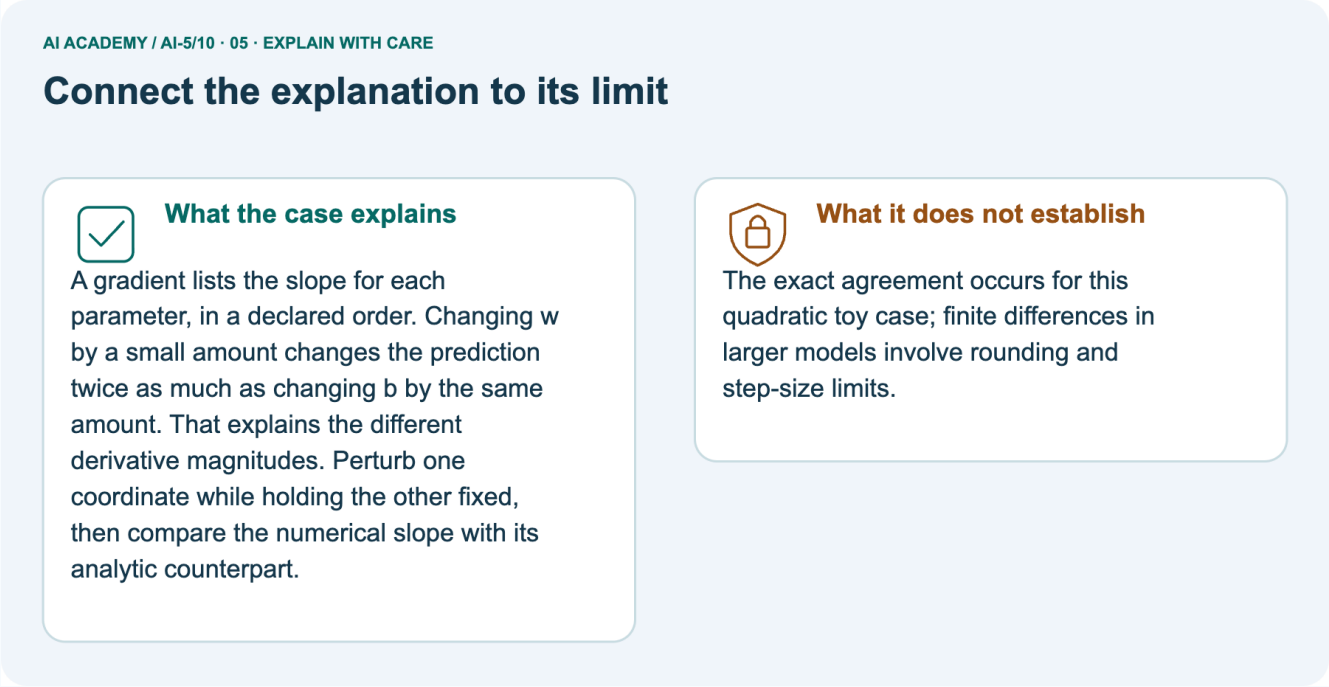
Original case: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. Changed condition: Swap the two **gradient** components in the program **output**.

Expected result: The coordinate-wise numerical check exposes the mismatch. Reason: The w perturbation changes y twice as strongly as the same b perturbation.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Limit: The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

> Explain the case, then point to a limit on the claim.

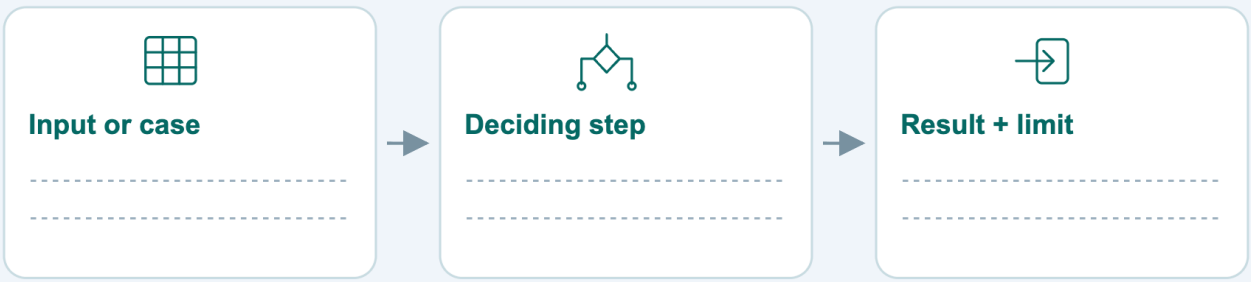
My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-5/10 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.